Assessing Public Awareness and Perceived Effectiveness of Sustainability Initiatives in Indian Railways

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Abstract

Indian Railways (IR) has committed to achieving net-zero carbon emissions by 2030 through large-scale electrification, renewable energy integration, and allied environmental programs. Yet the extent to which passengers are aware of, and perceive value in, these initiatives remains empirically underexplored. Descriptive demographic differences were observed in this exploratory, non-probability sample: respondents aged 23–27 reported the lowest awareness across all initiative categories, and awareness levels varied by gender across initiative themes. As the sample is small and non-randomly recruited, these patterns should be read as indicative rather than statistically established;, yet they nonetheless carry potential implications for IR's public communication and engagement strategy, pending confirmation in a larger more representative study.

Keywords: Indian Railways, sustainability, public awareness, perceived effectiveness, net-zero, electrification, demographic analysis

1. Introduction

Indian Railways (IR) has served as the backbone of India's mobility and logistics infrastructure since April 16, 1853, when the first passenger train traveled 34 kilometres from Bori Bunder to Thane under steam traction provided by three locomotives Sahib, Sultan, and Sindh (Indian Express, 2026). From that colonial-era origin, IR has evolved into the world's fourth-largest railway network, spanning approximately 70,000 route kilometres, transporting 7.15 billion passengers and 1.6 billion tonnes of freight annually (Press Information Bureau [PIB], 2026). Its scale renders it not merely a transport utility but a socio-economic engine, facilitating the movement of coal, food grains, and manufactured goods that underpin India's GDP.

IR's energy history mirrors its institutional evolution: from coal-fired steam in the nineteenth century, to diesel-electric traction in the twentieth, the network now stands at the threshold of its most consequential transition. The broader transport sector accounts for approximately 12% of India's greenhouse gas emissions, with railways contributing roughly 4% of that total (The Energy and Resources Institute [TERI], 2022). In response, IR has committed to becoming a net-zero carbon emitter by 2030 (PIB, 2023). The strategy rests on three pillars: achieving 100% broad-gauge electrification 99.4% complete as of early 2026, covering over 69,700 route kilometres, installing 30,000 MW of renewable energy capacity by 2030, and implementing systemic environmental programs spanning waste management, water conservation, and afforestation (Indian Infrastructure, 2026; PIB, 2023).

Despite this policy ambition, a critical knowledge gap persists in the literature. Existing scholarship has focused predominantly on technical and environmental performance indicators - kilometres electrified, carbon reduction estimates, and infrastructure targets achieved, while largely neglecting public awareness of and engagement with these initiatives (Banister, 2008). User perception is, however, a determinant of

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policy effectiveness: initiatives that remain invisible or poorly communicated to passengers cannot meaningfully shape travel behaviour or public support for green investment.

This study, published in collaboration with the NGO Hara Jeevan, addresses that gap. Its specific objectives are to: (i) quantify public awareness of key IR sustainability initiatives; (ii) evaluate the perceived effectiveness of those initiatives from a passenger perspective; (iii) analyse how demographic variables like age, gender, and educational attainment shape awareness and perception; and (iv) identify communication gaps to inform more effective public engagement strategies.

These objectives correspond to four guiding research questions: RQ1: How aware are IR passengers of specific sustainability initiatives? RQ2: How do passengers perceive the effectiveness of these initiatives? RQ3: How do age, gender, and educational attainment relate to awareness and perceived effectiveness? RQ4: What communication gaps can be identified between initiative visibility/impact and passenger awareness?

2. Literature Review

Sustainable transportation has emerged as a critical pillar of modern infrastructure planning, driven by its environmental, economic, and social implications. The transport sector is one of the foremost contributors to global greenhouse gas emissions, and the transition toward sustainable mobility has become central to international climate mitigation frameworks (Intergovernmental Panel on Climate Change [IPCC], 2022). Public transportation systems, particularly railways, are recognised as integral to reducing per-capita emissions, improving energy efficiency, and supporting sustainable urban development (United Nations, 2021).

Sustainability in transport is conventionally evaluated across four dimensions: environmental sustainability, economic viability, social inclusiveness, and technical or operational efficiency (Litman, 2020). Multi-criteria decision-making approaches, including the Analytical Hierarchy Process (AHP), have also been used in this literature to prioritise sustainability indicators (Saaty, 2008); the present study does not apply MCDM methods and instead reports descriptive awareness and perception data, as detailed in Section 3.

In the context of railway systems, the principal sustainability initiatives documented in the literature include electrification of rail networks, integration of renewable energy sources, deployment of waste management infrastructure, implementation of water conservation practices, and development of green station and corridor infrastructure. Electrification, in particular, has been established as the most impactful lever for emissions reduction, replacing diesel traction with electric locomotives powered increasingly by renewables (Britannica, 2023). Indian Railways has formalised this direction through its commitment to net-zero carbon emissions by 2030 (Ministry of Railways, 2023), supported by solar energy deployment at over 2,600 stations and the near-completion of broad-gauge electrification.

Allied programs have expanded IR's sustainability footprint. Waste management efforts include source-level segregation, composting, bio-methanation plants, and the nationwide deployment of bio-toilets in passenger coaches—a measure that has effectively eliminated direct discharge of human waste onto tracks, preventing both soil contamination and corrosion (PIB, 2022). Water conservation initiatives encompass over 2,400 rainwater harvesting installations and wastewater recycling systems at stations (Financial Express, 2021), alongside participation in Mission Amrit Sarovar, which supports the rejuvenation of local water bodies (India Today, 2025). Afforestation programs have seen IR plant approximately one crore trees annually since 2017, with 72 lakh saplings planted in 2021–22 alone (Ministry of Railways, 2022).

However, the existing body of research has concentrated almost exclusively on measurable environmental and technical outcomes, track kilometres electrified, tonnes of CO_2 avoided, or volume of waste processed. Considerably fewer studies examine public awareness of and engagement with these programs, or analyse how demographic variables shape passenger perception. This is a significant omission, as Banister (2008) argues that user perception is a key determinant of the social acceptance and long-term effectiveness of sustainability policy. Without understanding how passengers perceive and engage with these programs, policymakers lack the feedback loop necessary to optimise communication and deployment strategies.

This study addresses that gap directly, analysing public awareness and perceived effectiveness of IR sustainability initiatives through a structured survey instrument, with particular attention to demographic variation in awareness and perception.

3. Methodology

3.1 Research Design

This study employs a quantitative, cross-sectional survey design to evaluate public awareness and perception of sustainability initiatives implemented by Indian Railways. The methodological approach encompasses five analytical components: survey data collection, demographic profiling, awareness evaluation, perception and satisfaction analysis, and cross-demographic comparison. A quantitative design was selected given the study's objective of producing generalisable, comparable findings across demographic subgroups, and its suitability for identifying patterns in awareness and perception at scale.

The survey instrument was not formally validated (e.g., through expert review, cognitive pretesting, or a separate pilot study) prior to administration. Given the study's exploratory purpose, this is a limitation rather than a fatal flaw: the items asked about concrete, factual matters such as awareness of specific, named initiatives and straightforward ratings of perceived effectiveness, rather than abstract psychological constructs that typically require extensive validation to measure reliably. The data are accordingly best read as an indicative first look at the awareness landscape, with formal instrument validation recommended for any follow-up study intended to support stronger inferential or policy claims.

Formal reliability testing (e.g., Cronbach's alpha for the Likert-scale items, or test-retest reliability) was not conducted for this instrument. This reflects the exploratory, single-administration design of the study rather than an oversight specific to these items; a one-time survey of this kind does not directly support test-retest reliability, and the small number of Likert items limits the interpretability of an internal-consistency statistic even where technically computable. As with the instrument validation, formal reliability assessment is recommended for any subsequent, larger-scale study building on these findings.

No formal institutional or ethics-board review was sought for this survey. This is consistent with common practice for short, anonymous, non-sensitive opinion surveys of this kind and scale; respondent protection was nonetheless addressed through the informed-consent and anonymity measures described below, rather than through external ethics oversight.

3.2 Data Collection

A structured questionnaire was administered to 87 respondents representing a range of demographic profiles across age, gender, and educational attainment. The survey was distributed through educational and professional leadership networks, which accounts for the demographic skew observed in the sample,

notably the predominance of male respondents and postgraduate-level educational attainment. While a larger and more stratified sample would strengthen external validity, the 87-respondent sample spans eight distinct age cohorts - 15–22 (3 respondents), 23–27 (2 respondents) , 28-31 (0 respondents), 32–36 (3 respondents), 37–41 (8 respondents), 42-46 (35 respondents), 47-56 (27 respondents) and 56+ (9 respondents), however reported findings or data trends may be referenced to as broader categories for ease of representation, e.g. 18-33 or 37-55, enabling meaningful cross-demographic comparison within the scope of an exploratory study. The findings should accordingly be interpreted as indicative rather than nationally representative, and are best understood as a basis for generating hypotheses to be tested in subsequent, larger-scale research.

This constitutes non-probability, convenience sampling: respondents were recruited because they were reachable through the first author's professional and educational networks, not through random selection from the population of IR passengers. Recruitment was not restricted by gender, age, or education, and the resulting demographic skew reflects the composition of the sampling frame rather than deliberate selection.

The survey was administered online and distributed through social/professional communication channels to approximately 150 individuals, yielding 87 completed responses - a response rate of 58%. Eligibility was based on the respondent's membership in the professional and educational networks through which the survey was circulated; no additional screening criteria were applied beyond this network-based reach. Informed consent was obtained by notifying respondents, in the message accompanying the survey, that their responses would be used for research; a disclaimer at the top of the survey instrument reiterated that responses were anonymous and collected for research purposes. No personal identifying information was gathered, preserving respondent confidentiality, the only information collected was respondents age, gender, and education, for research and analysis purposes. The questionnaire was structured in three sequential parts:

(i) A demographic mapping section capturing age, gender, and educational background;

(ii) An awareness evaluation section assessing familiarity with specific IR sustainability initiatives; and

(iii) A perception section capturing respondents' assessments of initiative effectiveness, information sources, and observed environmental improvements. Likert-scale ratings, multiple-choice items, and open-ended questions were used in combination to capture both quantitative patterns and qualitative nuance.

Representative item wording was as follows. Awareness (Part ii) was assessed with: "Which of these sustainability initiatives of Indian Railways are you aware of?" (select all that apply; options: Track Electrification, Zero Carbon Emission Goal, Waste Management, Water Conservation, Tree Plantation to increase Carbon Sink, Dedicated Freight Corridors (DFCs), Green Stations/Infrastructure, plus an open "other" option).

Information source (Part iii) was assessed with: "How did you first learn about Indian Railways' sustainability initiatives?" (select all that apply; options: information at stations, on-train posters/announcements, news or social media, word of mouth, or "I was not aware").

Initiative-level perceived effectiveness (Part iii) was assessed with a single open-response item: "Which sustainability initiative do you think has been the most effective so far?"

Overall (not initiative-specific) effectiveness and satisfaction were each assessed on a 1–5 scale anchored "1 = negative" to "5 = positive," using the items "On a scale of 1–5, how effective do you think these

sustainability initiatives are?" and "Overall, how satisfied are you with Indian Railways' efforts toward sustainability?" respectively.

The primary independent (grouping) variables are age cohort, gender, and educational attainment, captured in Part (i). The primary dependent variables are: (a) awareness of each named sustainability initiative (categorical: aware / not aware), assessed in Part (ii); and (b) perceived effectiveness of each initiative (ordinal, Likert-scale), information source, and perceived environmental improvement, assessed in Part (iii).

Likert-scale items were scored from 1 (lowest rating) to 5 (best/highest rating). Multiple-choice items were of two types: opinion-based and perceived-effectiveness questions typically used a Yes / No / Maybe (Mostly) response set, while awareness-based questions allowed more than one option to be selected, with the specific options tailored to each question.

Two distinct question types underlie the effectiveness-related statements reported in Sections 4.3 and 5. First, two overall 1–5 Likert items ("how effective do you think these sustainability initiatives are" and "how satisfied are you with IR's efforts toward sustainability") captured a single aggregate rating across all initiatives combined, not a per-initiative score; for these, the count and percentage of respondents choosing each rating (1–5) is reported directly. Second, initiative-level statements such as "Track Electrification was the most frequently identified effective initiative" are derived from a separate, single open-response item asking respondents which specific initiative they considered most effective; free-text responses were grouped by initiative name (case-insensitive, with near-synonym responses such as "Electrification" and "Track Electrification" combined) before tallying, and the initiative receiving the largest combined share of responses is reported as most frequently identified.

3.3 Data Analysis

Descriptive statistics, frequency counts and percentage distributions were used for demographic profiling and initial awareness mapping. Cross-demographic comparisons were conducted across age cohorts, gender categories, and educational levels for all primary variables: awareness, perceived effectiveness, information sources, and environmental observations. Perceived effectiveness was assessed through Likert-scale scoring and aggregated satisfaction ratings. Recurring themes in open-ended responses were identified through manual thematic analysis.

Because the sample was recruited through non-probability convenience sampling and some subgroup sizes are small, cross-demographic comparisons are reported as descriptive differences observed within this sample rather than as statistically tested population-level effects, unless an inferential test is explicitly reported. Statistical analysis was conducted in Excel, which supports descriptive and cross-tabulated reporting but not inferential hypothesis testing.

4. Results

4.1 Demographic Profile

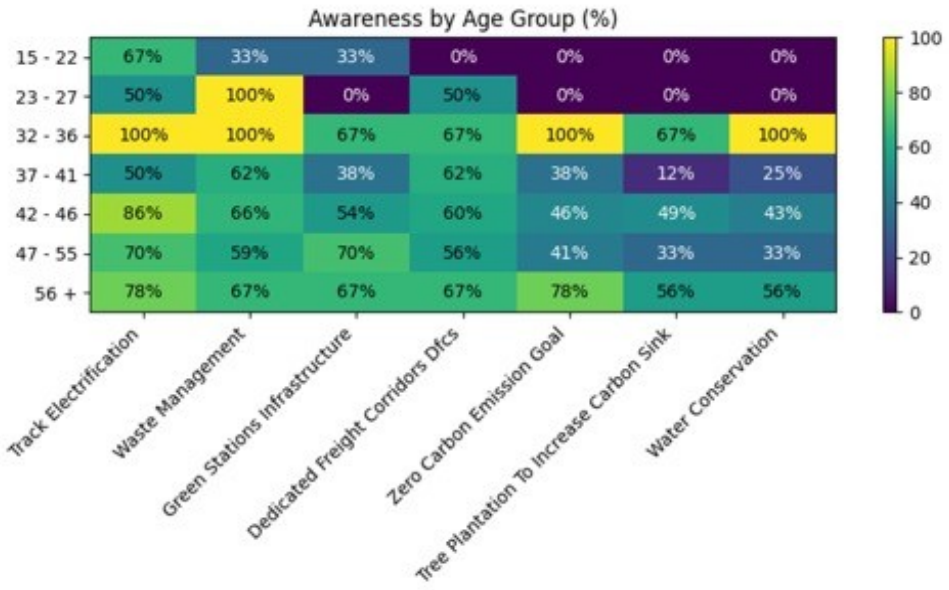
The survey sample was 70% male (61 respondents) and 30% female (26 respondents), reflecting the composition of the professional and educational networks through which the survey was distributed. The largest broad age cohort for representation purposes was 37–55 years (70 respondents total aged between 37-55). Educational attainment was distributed as follows: high school, 3.4% (3 respondents); graduate,

13.8% (12 respondents); postgraduate and above, 82.8% (72 respondents), indicating that the dataset represents a mature, highly educated segment of the railway-using population, one likely to have observed changes in railway infrastructure over an extended period of commuting experience.

4.2 Awareness of Sustainability Initiatives

Awareness varied considerably across initiatives and demographic subgroups, with one consistent finding: track electrification was the most widely recognised initiative across all groups, reflecting the inherent visibility of infrastructure-scale change.

Age-based awareness

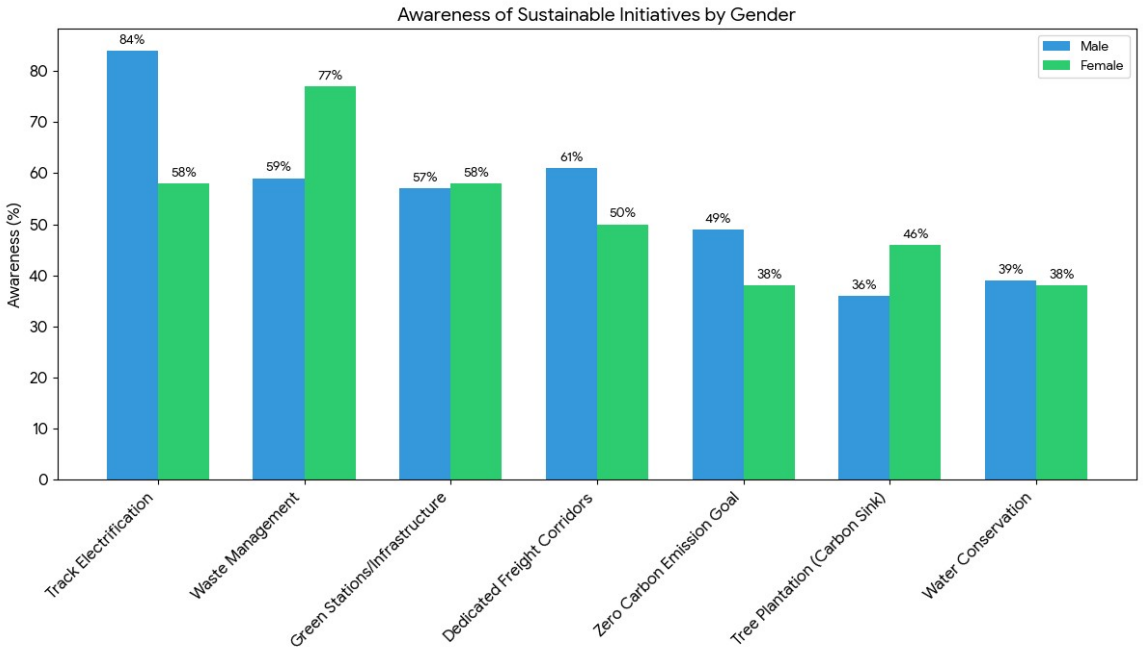


Source: Made by the Author

Respondents aged 32–36 (3.4% of the dataset - 3 respondents) demonstrated the highest awareness across nearly all initiatives, with 100% recognition of programs such as track electrification and waste management in both genders. Respondents aged 56 and above (9 respondents - 10.3%) also showed consistently high awareness, representing a demographic with strong potential for advocacy and storytelling-led outreach. The 37–55 cohort displayed uneven awareness across initiatives, suggesting selective engagement rather than systematic knowledge gaps.

The most significant finding in age-based analysis concerns the 23–27 (2 respondents, little representation obtained of this cohort) cohort, which recorded zero awareness of green station infrastructure, water conservation programs, zero carbon goals, and tree plantation initiatives. This is particularly concerning given that nearly 50% of India's railway-using population falls between the ages of 18 and 33 (Ministry of Statistics and Programme Implementation [MoSPI], 2025) the working-age demographic with the highest frequency of rail travel. The 15–22 cohort yielded insufficient responses for reliable analysis.

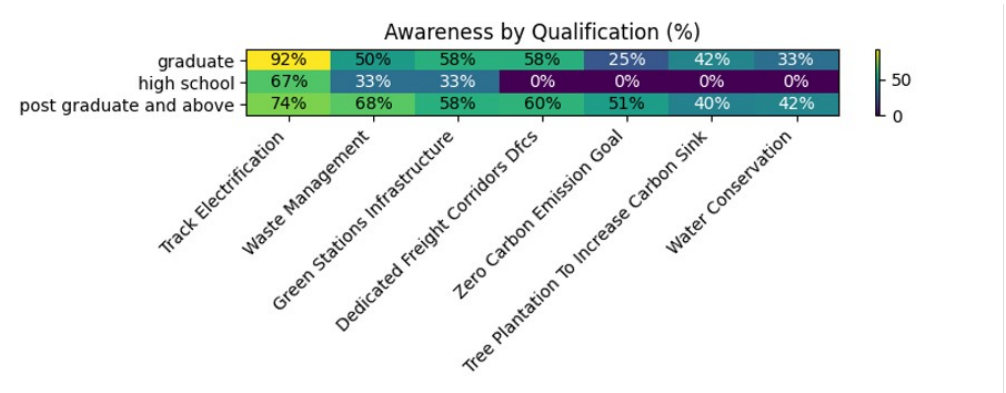
Gender-based awareness



Source: Made by the Author

A descriptive difference was observed between male and female respondents in this sample (n = 61 male, n = 26 female). Male respondents reported higher awareness of large-scale infrastructure and engineering initiatives: track electrification (84%, 51/61 of male respondents vs. 58%, 15/26 of female respondents) and Dedicated Freight Corridors (61%, 37/61 vs. 50%, 13/26). Female respondents reported greater awareness of initiatives affecting the immediate passenger environment: waste management (77%, 20/26 vs. 59%, 36/61) and tree plantation (46%, 12/26 vs. 36%, 22/61). These are associations specific to this sample rather than evidence of inherent or population-level gender differences, and should not be read as showing that men and women are generally predisposed toward different sustainability themes. An appropriate inferential test (e.g., chi-square) would be needed to establish whether these differences exceed what would be expected by chance.

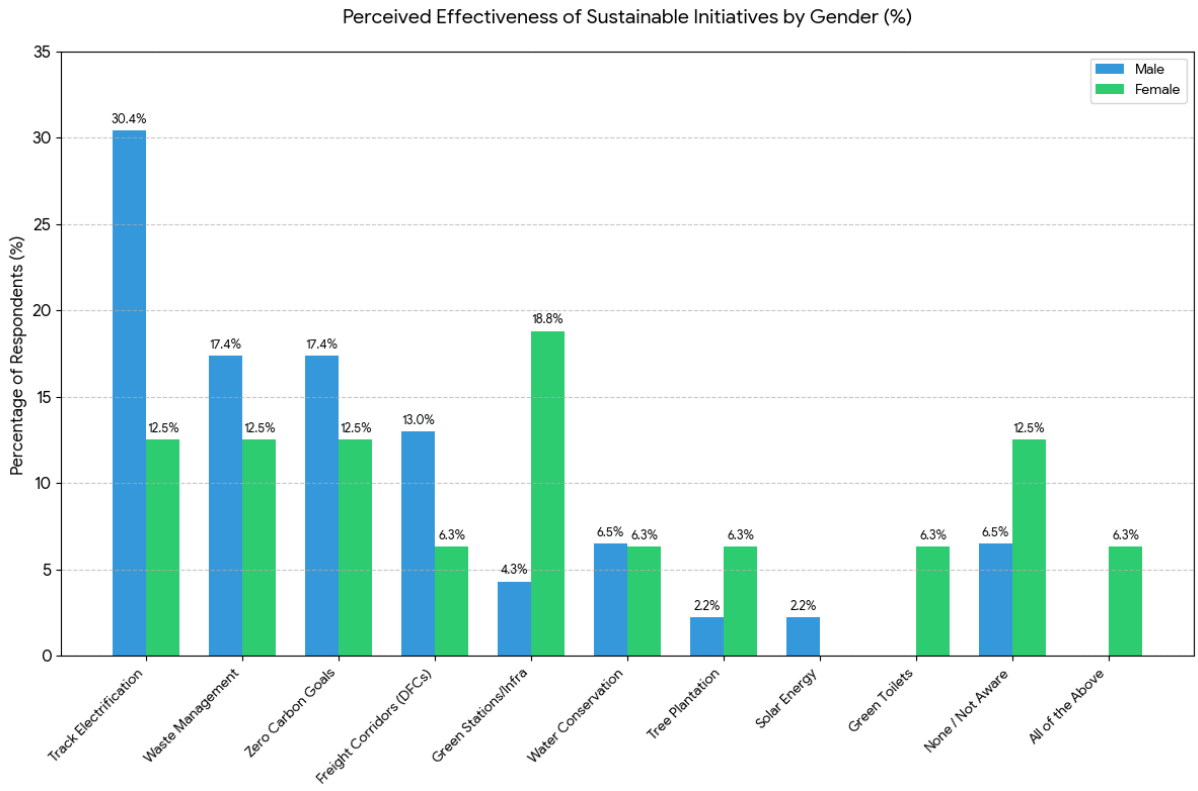
Education-based awareness



Source: Made by the Author

Undergraduate respondents (graduate category: 12 respondents) demonstrated the highest awareness of core infrastructure programs, including track electrification and Dedicated Freight Corridors. Postgraduate respondents (72) maintained moderate-to-high awareness with slightly more selective engagement. High school-educated respondents (3) recorded zero awareness of DFCs and carbon-neutral goals and showed the highest frequency of no-awareness responses overall. The limited number of high school respondents in the sample, however, warrants caution in interpreting these findings.

4.3 Perceived Effectiveness of Initiatives



Source: Made by Author

Track electrification was also rated as the most effective sustainability initiative overall, consistent with its dominant awareness profile.

Gender-based differences in perceived effectiveness followed a pattern similar to that observed in awareness, within this sample. Male respondents most frequently identified track electrification and Dedicated Freight Corridors as the most impactful initiatives. Female respondents most frequently cited waste management and green station infrastructure as the most effective programs. As with the awareness results above, these are associations observed in this particular, non-randomly sampled group rather than evidence of inherent or generalisable gender-based preferences, and should be interpreted with that caution.

4.4 Awareness Channels

Respondents identified news or social media as by far the most common source through which they first learned about IR's sustainability initiatives (67.1%, 57/85), followed by information displayed at railway stations (27.1%, 23/85), on-train posters/announcements (16.5%, 14/85), and word of mouth (15.3%, 13/85); 14.1% (12/85) reported having been unaware of the initiatives before the survey (2 responses were not obtained from total sample, total count for this section is 85 respondents). Age-differentiated trends were evident: older respondents reported learning primarily through word of mouth and official government communications, while younger respondents, despite being the demographic most dependent on digital channels, reported the lowest overall awareness, indicating that sustainability messaging is not effectively penetrating social media ecosystems where this cohort is most active.

4.5 Perceived Environmental Improvements

Only 39% of respondents (34 individuals) reported noticing definite positive environmental changes attributable to ongoing sustainability initiatives. A substantial proportion, approximately 35.5% (31 respondents) remained uncertain, while 24.5% (21 respondents) reported noticing no improvements at all. In aggregate, nearly 60% of respondents were either uncertain or unpersuaded of visible progress, one respondent failed to answer the question, hence the total respondents for this question are 86.

Among those who did perceive improvements, responses clustered around enhanced cleanliness, improved station facilities, waste management practices, and broader infrastructure modernisation including electrification. This pattern suggests that while sustainability programs are operational and producing measurable outcomes, their salience in the daily passenger experience remains limited or that passengers are not systematically attributing observable improvements to specific sustainability initiatives.

5. Case Study: Key Sustainability Initiatives

The material below combines externally sourced programme-performance data (e.g., published electrification, solar, and Dedicated Freight Corridor statistics) with findings from the present survey. Statements drawn from the survey are explicitly introduced as such (e.g., ‘survey findings indicate…’, ‘survey data suggests…’); all other statistics in this section are drawn from the secondary sources cited alongside them and were not measured as part of this study.

5.1 Track Electrification and Renewable Energy

IR has achieved near-complete electrification of its broad-gauge network, reaching 99.6% across more than 69,700 route kilometres as of early 2026 (Ministry of Railways, 2026). This transition has displaced diesel traction on broad-gauge routes at scale, yielding an estimated annual fuel cost saving of approximately USD 1.5 billion while reducing the railways' 4% contribution to India's national greenhouse gas emissions (Ministry of External Affairs, 2021).

To power this electrified network sustainably, IR's Green Railway Mission has installed solar panels at over 2,600 stations, with current solar capacity approaching 900 MW against an ultimate target of 20 GW (PIB, 2026). Approximately 70% of this solar energy is directed to traction purposes, with the remainder powering station operations. Survey findings indicate that graduates and postgraduates show high awareness of IR's zero carbon goals; however, the 18–33 age cohort expressed a desire for more visible

sustainability signals, suggesting that expanding station-level solar installations, rather than relying on remote solar farms, could improve public perception among younger passengers.

5.2 Waste Management and Bio-Toilets

One hundred percent of IR's passenger coaches are now fitted with anaerobic bio-toilets, effectively eliminating direct discharge of approximately 400,000 litres of human waste per day onto tracks. This has prevented significant soil contamination and reduced track corrosion costs by millions of rupees annually (PIB, 2022). Survey findings corroborate the importance of this initiative: 77% (20/26) of female respondents identified waste management as a high-priority sustainability area, making these visible, passenger-experience-oriented improvements critical for female passenger satisfaction and perception.

5.3 Water Conservation

IR has installed rainwater harvesting systems at over 2,400 locations and established wastewater recycling plants at stations for non-potable uses such as coach washing (Financial Express, 2021). Through participation in Mission Amrit Sarovar, IR has also supported the desilting and rejuvenation of local water bodies, with excavated soil repurposed for new track construction (India Today, 2025).

5.4 Dedicated Freight Corridors

The Dedicated Freight Corridor (DFC) project represents one of the most significant infrastructure investments in IR's history. By separating freight operations from passenger lines, DFCs reduce network congestion and enable trains to operate at higher speeds with heavier loads, including double-stack containers that carry twice the cargo of conventional configurations. On the Palanpur–Rewari stretch alone, the modal shift from road to rail saved 8.8 million litres of diesel and prevented 23 million kilograms of CO_2 emissions (Indian Infrastructure, 2026).

Despite this scale of environmental impact, primary survey data revealed awareness of DFCs at 58.8% across all demographic groups. This represents the study's communication failure: a flagship infrastructure project with measurable environmental benefits remains unknown to almost 40% of the passenger public, contributing negligibly to IR's public identity as a green operator.

5.5 Afforestation

IR has planted approximately one crore trees annually on vacant railway land since 2017, in collaboration with State Forest Departments responsible for plantation, maintenance, and disposal. In 2021–22 alone, 72 lakh saplings were planted (Ministry of Railways, 2022). Survey data suggests moderate awareness of this initiative, with female respondents showing marginally higher engagement than male respondents, , consistent with the pattern observed in Section 4.2 for this sample (rather than a general or population-level gender difference).

6. Discussion

6.1 Visibility as the Primary Driver of Awareness

The results consistently show that awareness tracks visibility: initiatives with a direct, observable presence in the passenger environment, electrification infrastructure, clean stations, bio-toilets command substantially higher recognition than technically impactful but operationally invisible programs such as

Dedicated Freight Corridors or afforestation. This finding aligns with Banister's (2008) argument that public perception of sustainability is mediated by the salience of changes in the immediate user environment, not merely by the scale of systemic intervention. IR's most consequential environmental achievements such as DFC emissions reductions, grid-scale solar generation, are precisely those least visible to the average passenger, creating a structural disconnect between impact and perception.

6.2 Demographic Gaps and Their Implications

The awareness gap among 23–27-year-olds is particularly consequential given that this cohort sits at the centre of India's most frequent railway-using demographic (MoSPI, 2025). Their near-zero awareness of green station programs, water conservation, zero-carbon goals, and afforestation points not to disinterest but to a failure of communication channels: sustainability messaging is not reaching the digital platforms and formats through which younger adults consume information. Conversely older respondents, particularly those aged 32–36 and 56 and above, show high awareness and could serve as effective ambassadors in community storytelling and advocacy campaigns.

Gender-differentiated awareness and perception patterns present a more nuanced communication challenge. Male and female respondents are effectively engaging with different dimensions of the same sustainability agenda, shaped by how each group experiences railway travel. Communication strategies that segment by thematic preference, technical infrastructure advancement for one audience, station-environment and hygiene improvements for another, are likely to be more effective than generic, undifferentiated messaging.

The education gap, while subject to the limitation of small high-school respondent numbers, suggests that accessible, non-technical communication formats are essential for reaching passengers without higher educational backgrounds, a group currently showing zero awareness of DFCs and carbon-neutral goals.

6.3 The Gap Between Initiatives and Perceived Impact

The finding that nearly 60% (52) of respondents were either uncertain or unpersuaded of visible environmental progress is significant. It does not imply that initiatives are ineffective as electrification and bio-toilet deployment data clearly demonstrate otherwise, but it does indicate that IR's communication strategy is failing to connect observable changes to their underlying sustainability causes. Passengers who notice cleaner stations or improved lighting may not attribute these changes to specific programs, and passengers who are unaware of DFCs cannot factor them into their assessment of IR's environmental credentials. Closing this attribution gap requires more direct, initiative-specific public communication.

7. Conclusion

This study examined public awareness and perceived effectiveness of Indian Railways' sustainability initiatives across a demographically varied survey sample. Three principal findings emerge. First, awareness is strongly mediated by the visibility of initiatives in the passenger environment: electrification and waste management are widely recognised; freight corridors and afforestation are not, despite their substantial environmental impact. Second, demographic variables (particularly age and gender) systematically shape both awareness and perception, producing identifiable communication gaps that current strategies are not adequately addressing. Third, the majority of respondents cannot confidently attribute observable environmental improvements to specific IR initiatives, indicating a disconnect between program delivery and public communication.

Based on this exploratory, 87-respondent, non-representative sample, several communication directions appear worth further testing: developing targeted digital communication strategies for younger passengers, piloting initiative-specific attribution messaging at stations (linking visible improvements to named programs), and exploring whether outreach segmented by demographic theme improves engagement. Given the sample's limited size and non-random recruitment, these are proposed as hypotheses for confirmation in a larger, more representative study rather than as established recommendations. High-impact, low-visibility programs may warrant dedicated public communication attention if broader research confirms the pattern observed here.

More broadly, achieving net-zero by 2030 is likely to require not only technical transformation but public awareness and buy-in. Within the limits of this exploratory sample, the study suggests that passengers' awareness of IR's environmental commitments does not yet match the scale of what has been implemented. Confirming whether this gap holds in a larger, representative sample would help clarify how much weight communication efforts should be given relative to continued technical investment.

8. Future Scope

Future research should expand the sample size and ensure balanced representation across gender, age cohort, and educational level, including finer age-band granularity (e.g., 56–60, 61–65) to replace the current collapsed 56+ category. Methodologically, the application of MCDM frameworks such as the Analytical Hierarchy Process would enable weighted prioritisation of sustainability initiatives by public importance, moving beyond descriptive percentages. Comparative analysis with other major railway systems, the Shinkansen in Japan, high-speed networks in Europe, would contextualise findings and identify globally transferable best practices. Finally, integrating hard environmental data (real-time carbon reduction metrics, water volume saved) with perception data would reveal whether current communication strategies are prioritising the initiatives that matter most, or whether high-impact, low-visibility programs require strategic repositioning in public-facing messaging.

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